

MAE 145: Planning and Estimation

Homework #1

Assigned April 2. Due on April 10

Notice:

Please upload several files for this homework assignment. In one file, please provide your answer to all of the problems, including a printout of your code with explanation in words/math on the logic that you used to derive your code. Provide the answers to the test cases in that file. In addition, please provide your code in a separate file.

Include in your written and electronic assignments the names of the other students you may have collaborated with to do the homework assignment. The Collaboration Policy for this course is detailed in the syllabus.

Homework exercises:

1. Exercise E1.2, from the lecture notes by Bullo and Smith.
2. Exercise E1.3, from the lecture notes by Bullo and Smith.
3. Exercise E1.6, from the lecture notes by Bullo and Smith.

Programming guidelines: Kindly follow the same syntax as shown below. If you do not follow the syntax your code will not work if it is tested.

The name of your python file should be `python_program.py` with the first line in the file having your PID as a comment, Ex: `# A432443`. Then,

```
p = [x, y]
```

```
q = [x, y]
```

where p and q are row vectors representing the x and y positions of the points.

Example:

```
p1 = [1, 1]
```

```
p2 = [2, 2]
```

```
q = [3, 4]
```

```
a, b, c = computeLineThroughTwoPoints(p1, p2)
```

where a, b and c are the parameters of the line defined as, $\{(x, y) | ax + by + c = 0\}$

```
distance = computeDistancePointToLine(q, p1, p2)
```

where distance is the distance of the point to the line.

```
distance, w = computeDistancePointToSegment(q, p1, p2)
```

where distance is the distance of the point to the segment, and w is 0, 1 or 2 as explained in E1.6.

For each of the functions above, provide logical explanations in comments of what your code is doing and why.

Then run your code and provide the output of each of the functions above for the following test cases:

- (a) Test 1: $a = -0.7071067811865475$, $b = 0.7071067811865475$, and $c : 0.0$
- (b) Test 2: $a = 0.0$, $b = 1.0$, and $c = -2.0$,
- (c) Test 3: $a = -1.0$, $b = 0.0$, and $c = 2.0$.